

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Branch: Mechanical & Allied Engineering

Semester: V

Subject Code & Name: BTAPE504D & Automobile Engineering

Max Marks: 60

Date:08-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define automobile and mention the detail classification of it with example.	CO1	6
B) What is a chassis? Explain with the diagram front engine rear wheel drive chassis layout?	CO2	6
C) Enlist the main components of an automobile and describe them in detail?	CO2	6
Q.2 Solve Any Two of the following.		12
A) I) What are the Functions of steering system?	CO1	3
II) Differentiate clearly between the functions of spring and shock absorber of a suspension system.		3
B) Explain with the help of diagram the steering linkage of independent suspension system?	CO2	6
C) List out the common trouble experienced in automobile suspension system and suggest the remedies in each case.	CO4	6
Q.3 Solve Any Two of the following.		12
A) Describe briefly with a neat sketch the construction and working of a single plate clutch.	CO3	6
B) Explain with a neat sketch the construction and working of a sliding mesh gearbox along with advantages and disadvantages?	CO2	6
C) Explain with a neat sketch the construction and working of a differential?	CO3	6
Q.4 Solve Any Two of the following.		12
A) What are the functions of brake? Classify it in detail with examples?	CO1	6
B) Draw a cross-section of automobile tyre and describe the basic constructional features of it?	CO2	6

C) With the help of neat sketch describe the operation of drum brake? CO3 6

Q. 5 Solve Any Two of the following. 12

A) Describe with neat sketch the construction and working of lead-acid battery used in automobile? CO3 6

B) Name the pollutants emitted by the IC engines in the atmosphere and mention its effects? CO5 6

C) Explain with a neat sketch the construction and working of magneto-coil ignition system in automobile? CO2 6

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